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Q1. (a) What is a negative voltage?

(b) What is the difference between +48V and -48V?

(c) Why is BSNL using -48V in all its equipment?

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A1. (a) The voltage at a node is measured in volts. It is the potential difference between two points or the difference in electric potential energy per unit charge between two points.

The voltage at a node is measured with respect to another point which is taken as reference. The reference point in most cases is with respect to a common node known as ground. When we say that the voltage at node 1 is 5V, it means that the voltage at node 1 is higher than ground by 5V. When we connect another resistor R between the node and ground, assuming that the voltage is maintained at 5V, current in the resistor is equal to $5/R$ as per Ohm's Law.

The negative voltage at a node means that it is at a lower voltage than reference to ground.

(b) Difference between +48V and -48V can be explained with the help of an example. Let's say that the voltage at node 1 is +48V and node 2 is -48V. These voltages are measured with respect to a common node referred to as ground. That is, node 1 is 48V higher than ground and node 2 is lower than ground by 48V.

The voltage measured with node 2 as reference will be 96V at node 1 and 48V at the erstwhile ground. There is no minus sign appearing with the voltage. So we conclude that plus or minus voltage only indicates the level of the voltage with a reference point.

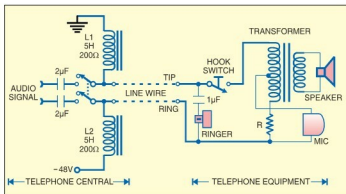
For better understanding, take the case of the first floor, second floor and third floor in a multistoried building. The first floor is lower than the second floor, and the third floor is higher than the second floor.

(c) The current which powers your telephone is generated from the 48V bus in the central exchange. 48V is sent to the telephone line through some resistors and inductors in series with the 48V power source.

The figure here shows the typical arrangement of a traditional telephone line interface. -48V is sent by the central telephone exchange through the Tip and Ring pair of lines. The 48V level is enough to sustain the voltage level along the line.

Some of the reasons for using -48V in the telephone lines are:

1. Low voltages are safer for telecom technicians and service personnel.
2. 48V is sufficient for transmitting



Circuit of traditional telephone line interface

power through telephone lines to a long distance.

3. A negative line voltage helps to reduce the corrosion from electrochemical reactions occurring on a wet telephone line.

Q2. Compared to other devices, why MOSFET is so important in the VLSI design technology?

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A2. There are two types of transistors—bipolar and metal-oxide semiconductor (MOS). The basic difference between these transistors is in the